

Database Constraints

Pets, owners, and rules for their relationship

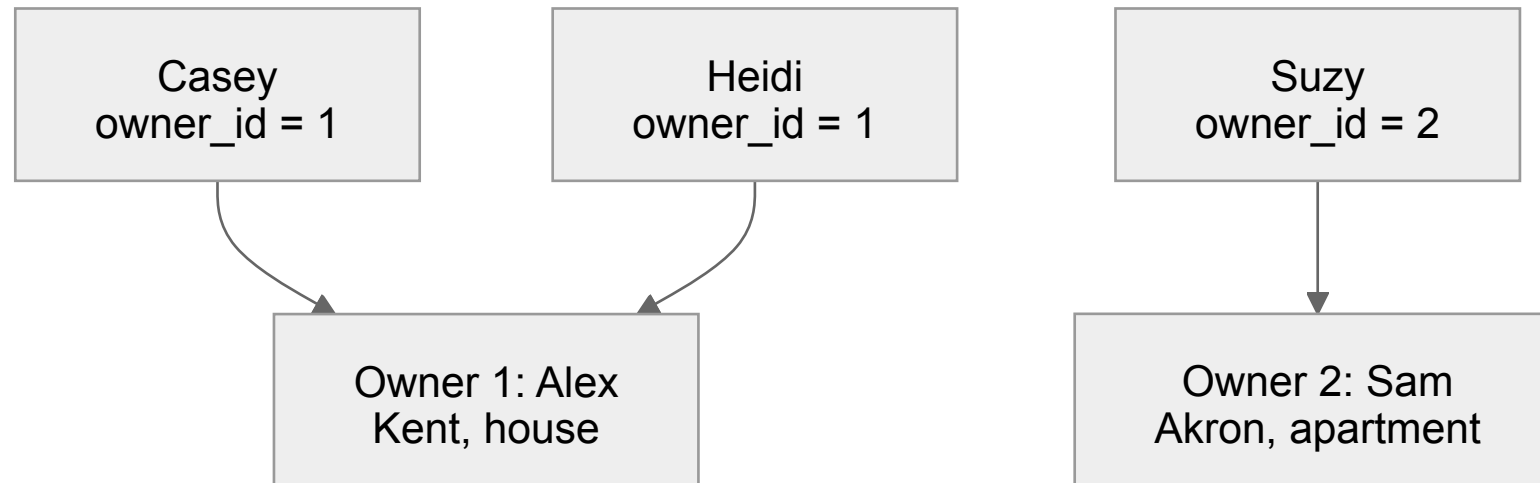
Gregory S. DeLozier, PhD

One Owner, Several Pets

Pet	Owner	City
Casey	Alex	Kent
Heidi	Alex	Kent

Alex moves to Akron. Where should that change live?

References to Owner Records



One owner can have many pets. Each pet has one owner.

Running the Example

```
python3 setup_database.py  
flask run
```

`http://127.0.0.1:5000/list`

Manage Owners opens the owner pages.

The Owner Table

```
CREATE TABLE owner (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    name TEXT NOT NULL,  
    city TEXT,  
    type_of_home TEXT  
);
```

id identifies the person. name describes them.

The Pet Table

```
CREATE TABLE pet (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    name TEXT NOT NULL,  
    type TEXT NOT NULL,  
    age INTEGER,  
    food TEXT,  
    owner_id INTEGER NOT NULL,  
    FOREIGN KEY (owner_id) REFERENCES owner(id)  
        ON DELETE RESTRICT  
);
```

Two Rules on owner_id

Value	Result
NULL	Rejected by NOT NULL
An absent owner's identifier	Rejected by the foreign key
An existing owner's identifier	Accepted

Referential integrity: references point to existing records.

Foreign Keys on This Connection

```
connection.execute("PRAGMA foreign_keys = ON")

fk = connection.execute(
    "PRAGMA foreign_keys"
).fetchone()[0]
assert fk == 1
```

A separate connection needs its own setting.

The Database Interface

Pet operation

```
get_pets()
```

```
get_pet(id)
```

```
create_pet(data)
```

```
update_pet(id, data)
```

```
delete_pet(id)
```

Owner operation

```
get_owners()
```

```
get_owner(id)
```

```
create_owner(data)
```

```
update_owner(id, data)
```

```
delete_owner(id)
```

Names on Screen, Identifiers in the Form

```
<select name="owner_id">
  {% for owner in owners %}
    <option value="{{ owner['id'] }}">
      {{ owner['name'] }}
    </option>
  {% endfor %}
</select>
```

Alex appears on screen. "1" travels with the request.

Saving the Reference

```
cursor.execute(  
    "INSERT INTO pet(name, age, type, food, owner_id) "  
    "VALUES (?, ?, ?, ?, ?)",  
    (data["name"], data["age"], data["type"],  
     data.get("food", ""), data["owner_id"]),  
)  
connection.commit()
```

SQLite checks the relationship during the insert.

Application Checks and Constraints

Check	Where it happens
Owner selection is blank	Flask route
Owner selection contains non-digits	Flask route
Referenced owner exists	SQLite
Required fields are not NULL	SQLite

A dropdown helps selection. The database checks the relationship.

Displaying Owner Names

```
for pet in pets:
    pet["owner_name"] = "<Unknown>"
    for owner in owners:
        if owner["id"] == pet["owner_id"]:
            pet["owner_name"] = owner["name"]
```

owner_name is for display. owner_id is stored.

Reassigning a Pet

Select **Sam** in Casey's owner dropdown, then save.

```
database.update_pet(10, {
    "name": "Casey", "age": "9", "type": "dog",
    "food": "kibble", "owner_id": "2"
})
```

Pet	Before	After
Casey, id=10	Alex, owner_id=1	Sam, owner_id=2

Casey's other fields stay the same. Both owners remain.

Deleting an Owner

```
FOREIGN KEY (owner_id) REFERENCES owner(id)  
ON DELETE RESTRICT
```

Alex still has pets: deletion fails.

Reassign those pets, then delete Alex: deletion succeeds.

Testing the Relationship

```
python3 database.py
```

Test operation	Expected result
Create pet with absent owner	Constraint error
Delete owner who has pets	Constraint error
Remove pet, then delete owner	Success

Food as a Shared Record

Two pets eat the same food.

What would change if food had its own table?

Now

`pet.food` contains text

Form accepts a food name

Proposed exercise

`pet.food_id` refers to a record

Form selects an existing food

Rules Belong with the Records

Store the identifier that defines the relationship.

Let the database enforce whether that reference is valid.

Test both accepted changes and rejected changes.